Susan Lewis **Essay 3**

ENEP 821 **Fall 2012**

The relationship of technology, environment, and society is a complex and varied. As science and technology have progressed, so too has social complexity and economic growth dependency. The environment is the most fundamental, essential, and important piece of the puzzle, but sadly not treated as such. Society – life - is completely dependent on the environment, whether because it is valued as our home or because it is valued for its economic or resource worth, society is still dependent on it. The relationship of technology and environment is one of duality. The rewards of grand technological advances have come with grave environmental and social risks. Technology has enabled society to understand and scientifically evaluate the environment, for instance, understanding the atmosphere and the changes humans have caused to its chemistry (Byrne, et al, 2002:282). The cause of the changing chemistry and resultant climate change, though, is science and technology.

On the macro level, society is lacking appropriate balance with environment and technology. The environment should be our number priority and point of respect and balance. Without a healthy environment, we cannot have a healthy society or population. Technology can only replace and attempt to replicate or engineer so much; and that limitation has been forgotten or lost. The environment and society do not exist in isolation from one another. We sometimes use technology to force separation, but it is not real. Science and technology are prioritized over nature and society (Shiva, 1992: 21) and this is an inappropriate relationship, to say the least. Beyond inappropriate, it is becoming an increasingly dangerous.

The root of this issue stems from far before technology was a major player in environmental or societal injustice and deterioration. The root cause of the imbalance and mistaken priorities is a philosophical issue among much of the

human race. With the rise of the science and technology revolutions and capitalism (Byrne, et al, 2002:286), priorities and Western mentality became focused on development, economics, and commodification of everything possible. With this came a drastic change in the relationship between society and environment. The goal now is perpetual growth and the natural world is treated as a commodity, as a resource to be used, not valued for itself or as a global commons. While there is a list of social-environmental injustices, imbalances, and problems to solve, we face a much deeper-rooted issue than this list. It is difficult to say what is needed to fix all that is broken socially and environment. It is also difficult to avoid becoming pessimistic and best and fatalistic at worst in considering the depth of change needed to get to the root of the problems.

1) One might perceive the root of social-environmental problems as the point in time where the concept of "environmental crisis" was given its name and definition. More accurately though, is that this point in history is when society became aware of large-scale problems between society and environment, not where the problems actually came into existence nor the true root of them. The conceptual seed was planted for society but the path leading to this place was paved long before. At this time, societal concern began to grow about resource use and pollution. There were some concrete, real-world illustrations of the problem at hand, which is part of what led to new goal of sustainable development. It was a new way of thinking about the world and how society and environment fit together. Rachel Carson gave the environment its first voice with Silent Spring in 1962. It was at this time that society and technology had an identifiable negative impact on the environment, prime examples being use of DDT and the first major oil spills.

There was a shift at this point in time – it was not the sudden birth of social-environmental problems, but it was the birth of widespread social awareness about the problems. With that came new terminology, like biodiversity and endangered species. There was a new major crisis to solve and out of that

demand was born the loose idea of sustainable development and a goal to achieve it, once we could figure out what it actually meant. The WCED (1987:1) designed the first working definition of sustainable development, and it was conceptualized and expected to fix both the environmental crisis and the growing poverty crisis. Unfortunately, it did neither. This is not a criticism of the ideas or words from the WCED – the barriers to repairing what is broken are larger scale and more deeply rooted than what a team of experts can solve in a short term. The social-environmental problems are more than a list of needed repairs, but a philosophical issue of how humans perceive the natural environment.

The World Commission on Environment and Development was established and charged with solving the social-environmental problems. The WCED sought to find a way to *manage* natural resources to sustain human health and progress. Environmental issues like acid rain and fossil fuel consumption, and social issues like starvation and education needed solutions. Economic development and environmental issues were to be integrated in management, and a broader, cooperative, international perspective adopted (WCED, 1987:2). Environment, development, and energy were identified as one connected crisis and both the industrialized and developing world were to have the same new goal of sustainable development (WCED, 1987:3). While the integrative ideas are sensible and the intent was for progress and equality for all people and the natural world, the goals have not been achieved. The crises persist. The fundamental philosophical problem was evident here too – in that the natural world requires human management (Sachs, 1993:18). This is a problem with sustainable development in concept and definition.

Further challenging is that sustainable development is synonymous with sustainable growth. Sustainable development, as a process of change, could work but only if it can exist without continued growth (Daly, 1990:45). This is a fundamental problem with the idea of sustainable growth. It is impossible to have successful sustainable growth because we live in a world with finite limits (Daly, 1990:45). Growth, economic or otherwise, requires natural capital and the Earth

can only tolerate so much. The planet can only assimilate so much pollution and waste, and humans can only take so much natural resource out of nature without creating an unsustainable situation (Daly, 1990:46). We likely have already done so. There is a serious conflict here and definitional barrier to sustainable growth.

There seems to exist a duality in world-view in the West, as if society and environment exist separately, in distinction or isolation from one another (Byrne, et al, 2002: 264). In the social world, the idea for sustainable growth focuses on the growth piece –continuous economic growth and economy building, and views nature as a resource under our control. It is thought that economic growth is the only direction to go, the proposed and expected solution to problems. The environmental side of the equation does not allow continuous growth because of natural finite limitations that cannot be overcome. Further, environmental limitations are treated as constraints and it is conceptualized that through technology, we can overcome them. It simply is not true. Climate change is perhaps the ultimate environmental constraint. It pains me to think we might one day attempt to "overcome" climate through technological means. The solution is in changing both our ways and perceptions of the natural world, not managing the operations of the natural world. It is a problematic perspective that natural limitations of Earth are potentially overcome with the right technology and tools. Problematic also is the commodification of natural resources. Production and economic growth have been prioritized to the point of catastrophic ecological degradation (O'Connor, 1989:5). Society and nature are not separate phenomena, no matter how much we behave as if it were or how hard we try to force it to be so.

The perhaps best-known illustration of this phenomenon and the rise of social-environmental problems is with the Green Revolution (Shiva, 1992 & O'Connor, 1989). This is an example of developmental pursuits driven by the technology-focused West, resulting in the detriment of both societies and environment. It was based on the belief that technology is superior to nature. The Green Revolution was an attempt to use technology to make the

environment, specifically agriculture, work better. It was offered as a "miracle recipe for prosperity" (Shiva, 1992:20). The developed West overpowered developing societies, transferring science, technology, and means of production and consumption. Also transferred were reductionist agriculture, pollution, and dangerous chemicals (O'Connor, 1989:10). All of this under the auspice of development, because Western ways – based in technology and science - were superior. Science in particular was placed above all, including society (Shiva, 1992: 21). Ultimately, the necessity of fertilizer, pesticides, irrigation and socio-ecological costs of the Green Revolution did not help uneven development or starvation problems as it was intended to – quite the opposite. The mid and small scale farmers could not compete with the subsidized, large-scale agriculture brought in by the West. The society was arguably robbed - of diversity, culture, knowledge, and land. Also taken from the indigenous culture something central to their agricultural methods and policy - working in partnership with natural processes (Shiva, 1992:29). I can only imagine what the world might look like if we had learned from them instead of taking from them.

The true root of social-environmental problems precedes the Green Revolution, sustainable development, and the environmental crisis. Norgaard argues that the root stems much farther back into history, even before the industrial revolution. He describes a need to review, reclaim, and understand our development history and a need to look at the process, not only the product (Norgaard, 1994:32). Our history is that of viewing and using natural resources without regard to limits or negative environmental or social impacts. That production and development are more important than equality and nature. That human greed has led to ignoring nature and exploiting the poor (O'Connor, 1989:9). That humans are superior to nature and need to manage it. Now we find ourselves in a world of duality between society and environment. Development is so deeply engrained in Western mentality that it has become a way of thinking. We persist with goals of development, despite failure (Sachs, 1993:9). This was evident even at the UN 'Earth Summit' – in that the approach to solving the environmental crisis would be based in development. The summit

“inaugurated environmentalism as the highest state of developmentalism” (Sachs, 1993:3). There are deep seeded beliefs and philosophies in the West that have led us to the social-environmental problems faced today.

This history had led to the concept that perpetuating and accelerating development and technological change is the solution to unsustainability (Norgaard, 1994: 34) and social-environmental problems. It is contradictory and brought us to the slippery slope of technicism. Until society can rectify the misconceptions, false beliefs, and dangerous addiction to development and technological solutions, we cannot truly move forward into sustainability. It is more than a list of problems to solve, but a philosophical perception of the world.

An important missing piece in society is a metanarrative. There is a need for a common goal, a constructive shared vision for the future (Norgaard, 1994:185). Post modernism does not allow for that and a significant shift is needed to move away from fragmentation and disconnection between society and environment and between each other, and instead towards living in reality where society and environment, people and nature, are not separate phenomena. In the process and rise of the technological transition, society disconnected from the environment. People developed a sense of freedom from the environment and of having control over nature, an ability to overcome environmental constraints, and enabling them to design their future as they wish (Norgaard, 1994:45). This is a huge misconception and mistake to pursue. We are not separate from our environment, it is our home; we are part of it and it is part of us. We are also dependent upon it, and that too seems to be lost in technology. The world of modernity and Western methods is not sustainable, and the developmental and technological based solutions to our environmental and social problems are not working. “Experts” cannot solve the problems, with or without technology. We need to recognize and hold ourselves accountable for making poor decisions in our environmental history and re-evaluate and redeisgn our philosophy.

A fundamental issue with development is: who/where really needs development? Sustainable development needs to begin with the industrialized countries (Daly, 1990:46). This is where a change in how natural resources, energy, food supply, etc are used. The social/environmental philosophy and history in the West needs to be managed, not the natural world. Instead of focusing so much attention on development and getting the developing world up to Western standards, we need to address what Western standard are and why. Social-environmental problems can be traced back to the developed and industrialized world and the world-view that has been created over time. It will not be easy as the ideas are deeply engrained in Western mentality, culture, and history. Nonetheless, the solutions are not in the form of technology but in the form of changing how we exist.

2) The future of the commons regimes is approaching a dangerous tipping point. The dualism of society creates a significant challenge to commonality. Technology and science are placed above all else, including society (Shiva, 1992: 21) and we are quickly moving towards placing it as superior to the most elemental and sacred parts of our existence. Coming out of the disconnected, fragmented nature of post-modernism, we need to find connection, commonality. There is perhaps little left in our world that might be considered the commons, but it is some of the most precious and pure aspects of the human experience.

It is hard to have a clear view for the future of the commons regimes when the definition of the commons is ambiguous. My perception of the commons is that which inherently belongs to or is accessible to all people. Things that everyone has fair rights to. The commons are things like the atmosphere and ocean, the beauty of the aurora borealis, the biodiversity of coral reefs, and the mystery and awe of space. I think the commons could arguably include food and human survival – basic, intrinsic, and necessary for life. Also included should be values and morals, and perhaps even artistic expression. Commons are parts of life, reality, and our world that make us feel alive and have incalculable value

simply by virtue of what they are. These are elements of life that are not manufactured, sold, franchised, or controlled by society or technology. These are things that should not be technicized or broken down into fragments. They are not puzzles or problems to be solved; they are commonalities among all of us, everywhere. Further clouding the view of the commons regimes future is its intimate relationship with science and technology. If there is a grim future ahead, it will be a result of imbalance between technical knowledge, its application, and maintaining basic values (Flyvbjerg, 2001:2)

I would describe the commons also as that which we have not yet taken control of, figured out how to breakdown, sell, or make a better version of through technology. The commons, like the natural world, do not need to be managed. They should not be altered or reconstructed. Contrary to some belief, human reason is not superior to nature's design. Society does not understand nature intricately enough that we might improve it by our own means (Byrne, et al, 2002:262). That said, we are approaching a dangerous place where the fundamental respect that should be given to the ecological commons is breaking down. The science and technology focused and dependent West is working on ways to control the commons, even commodify aspects of it (Byrne, 2002: 270). Human arrogance rears its ugly head again, in the worst way yet. Science is dangerously placed as more important and superior to all things, including nature and society. Even something as pure and biological as human reproduction is being evaluated scientifically for more efficient strategies (Shiva, 1994:139). We are losing what should be unbreakable value and respect for basic human reproduction in the name of efficiency. We risk altering our self-definition by establishing false superiority of human creative design over all things.

If we commodify the commons or attempt to redesign it in a way that is superior to nature (Byrne, et al, 2002:265) we are sure to greatly diminish inherent value in our world at best and tragically fail at worst. We have already destroyed natural beauty on Earth through greed and an endless need to produce and consume. If we continue this way unabated, if we attempt to control

or manage nature further - the atmosphere, the ocean, and space - we run the risk of literal self-destruction. We can already see evidence of tampering with nature and the commons – in this case the atmosphere - in the form of climate change (Byrne, et al, 2002:265). I would not put it past humans to keep going down this path until we kill ourselves. Even if not literal self destruction, devaluing the commons and redesigning our reality in our own terms could leave us in world without meaning, without soul, without wonder and astonishment. It is a mistake to attempt to turn all that is natural and deeply inspiring into a constructed reality of human design (Shiva, 1994:141).

If we lose basic appreciation for nature and biology, for each other, for beauty that simply is, what do we have left? Where do we go to find center and get back to our roots? Where would we find tranquility? This cannot be manufactured. The commons provides these precious aspects of existence because of something basic, intrinsic, and raw in the human/environmental connection. There is a reason why so much of the population lives within close proximity to a coastline, for instance. Many of us feel an inexplicable sense of calm and peace when standing near vast bodies of water. Human bodies are 75% water and so is our planet. We are connected in ways that cannot and should not be broken down into puzzle pieces and re-arranged. There is a harmonious and vital connection between the human spirit and the natural world, between humans and animals, something inexplicable in the awe of majestic mountains, the tranquility of a calm coastline, and the sense of wonder about outer space. Losing this would be a true tragedy of the commons.

Technology itself is a dichotomy, a duality - on one hand, it has enabled great advances like medicine, communication, and space travel, to name a few. It has allowed information transfer, global communication and connections previously unimaginable. It has also become a force of its own and is perceived as the answer to all problems, even those created by technology in the first place (Byrne, 2002:271). We risk getting lost in the enchantment of technology because of a mindset that technological solutions or systems will be somehow

better or more efficient that natural ones. Technology connects and separates us at the same time. We are more connected than ever, yet increasingly disconnected from nature and the commons.

Society today has the dangerous ability to disconnect, alienate, to be separate, from itself, from nature, from commonality. The separation enables the vision of being able to create something better by undoing the commons and putting it back together again in our vision (Byrne, et al, 2002:265). We have already undone the commons of food, agriculture, seeds and natural reproduction through the Green Revolution and forms of bioengineering. Where does it end? We have begun mechanizing childbirth – is there anything more sacred and pure, and common to all human beings? Humans are edging close to engineering themselves (Shiva, 1994:138). Human arrogance has really hit a new pinnacle if we think we can reproduce ourselves better or more efficiently than nature designed.

All of that said, the future of the commons regimes is not definite or absolute. Humans are not determined for self destruction, but it is a real possibility if we do not slow down and re-evaluate what is important and why. There is nothing wrong with technology and science, only in what we do with it. We need to balance technical knowledge with understanding of values; right and wrong (Flyvbjerg, 2001:2). Humans do not need to manage all aspects of life and earth. There are aspects of existence that make life the amazing, awe-inspiring, wondrous thing that it is. I believe that somewhere, deep down, every one of us can still tap into that natural connection to our commons. For some it might be harder to find than for others, but we all have it because its human nature. Perhaps better stated as human commons, in this context. There is still hope for the future of the commons, we have not yet completely destroyed or reconstructed all of our commonalities. It is a real danger, though, and we must look within and around ourselves before we go over the edge.

3) The relationship of social inquiry, technology, and environment is complex and eternally intertwined. It is also in a state of serious disrepair. Over the course of evolution and history, humans – along with all other life in the biosphere – grew and evolved with the natural world. They were not separate entities but existed as one, as life on earth. The natural world was society's teacher, instructing what was real or not, was possible or not. Things are different now; the rise of scientific knowledge coupled with a capitalism, characterized by growing obsession with production, consumption, and money marked a point of identifiable change in societal priorities and thus relationship with the environment. Human progress became synonymous with the mastery of nature (Byrne, 2002:261), a far cry from peaceful harmonious coexistence. Natural resources have become just that – resources - for our use and disposal. Science and technology allowed society great access and ability to overcome pesky natural constraints, like those dead dinosaurs and their hydrocarbons buried so far under the ground.

Technology and the technological/knowledge revolution created an interesting dualistic relationship with social inquiry and environment. Social inquiry, curiosity, desire to grow, and the need to control are what paved the way to the technological transition and to becoming a technological society. This too, appears to be out of balance, illustrated by the greatest scientific and technological accomplishments bringing the greatest risk and destruction to society and environment. This can be seen in situations of oil spills and nuclear disasters. Ships that carry oil are sophisticated, complex machines, which could not be designed or operated without modern science and technology. Accidents with these scientific marvels bring equally unprecedented death and destruction (Byrne, et al, 2002: 273). Nuclear fission, and thus power, represents scientific understanding and application of the highest caliber. We also now have the power to completely destroy ourselves – in both society and nature. The important question is, are we smart enough not to? We are beginning to see this

same dualism in climate change and alteration of atmospheric chemistry. The difference, for now, is that the former examples led to extreme, identifiable tragedy in the form of human disease and death. Climate change cannot yet be so definitively linked to death and destruction, and I believe this is part of why it is so difficult society to make adjustments and changes. Sadly, society today tends to remain passive, indifferent, apathetic, and disconnected to situations until they are forced to participate, contribute, or at least pay attention.

The relationship between social inquiry, technology, and environment is out of balance. Technology and science have been given higher value and priority over society and environment (Shiva, 1992: 21). Contrary to some belief, technology coupled with human reason cannot create a better, more efficient version of nature (Byrne, et al, 2002:262). Nor can they manufacture or engineer replacements, particularly of the commons, for what humans destroy. Technology has power and value and the environment does not. Society has become passive and looks to technology for solutions, largely regardless of environmental implications. It is clear that the relationship between social inquiry, technology, and environment is greatly imbalanced and priorities skewed.

The question of the day, and of our future, is what the relationship ought to be. There should be balance and correct prioritization. The environment is the ultimate commons; it is home. We would not destroy the house we live in nor should we destroy the planet that we collectively live in. Environment should be prioritized over technology and science, not the other way around. Technology should be a tool, an instrument for society to grow in a healthy, balanced, and intelligent way. Technology has been given ultimate power- it leads us, guides us, and we mistakenly see it as the solution to all problems. Social inquiry towards technological progress does not need to end, nor would it, realistically. Technology and applications need to remain under control and in balance with global priorities and equality, though, and not be made into some kind of demigod.

The appropriate relationship of society and nature is not separation and isolation, but connection and solidarity. There is no real separation between humans and nature and we must stop behaving as if there is. Technological progress, social progress, social equality, and knowledge should emerge and evolve through connection and partnership (Shiva 1994:288), in harmony with the environment. Society and nature are not passive operations. Technology, if any, is passive and controllable. We created it. Technology and science should help us, not define us. Again – the priorities and balance need realignment. Further in need of balance is social inquiry as related to science and technology. Humans are creative, curious, and capable of producing amazing things. Technology and technological progress are not inherently bad or evil. The trouble is when applications of technology are poor or shortsighted, creating severe negative impacts. There needs to be a balance between pursuit of technical knowledge and maintaining basic value and respect for the environment and each other (Flyvbjerg, 2001:2).

Whether or not this vision of balance and corrected priorities can actually become a reality is not certain. The way society exists with technology and environment is deeply engrained as a way of life, arguably a philosophy, and will be very difficult to change. It will be particularly difficult to change in the short period of time that we have left before going over a dangerous technological cliff.

At the risk of sounding pessimistic or fatalistic, I think the necessary changes to truly achieve global sustainability may come out of catastrophe. Historically, society "gets it" and makes adjustments not preventatively, but when absolutely forced. Humans are very intelligent beings, but not more intelligent than nature. Human arrogance is strong as ever and humility is sadly missing. We are headed down a dangerous path and rapidly approaching a point of no safe return. The relationship between society, environment, and technology is abusive and in dire disrepair. It is up to us to fix it if we are to have a long, harmonious existence on Earth. It is also ours to destroy. The ultimate

challenge today is to rediscover commonality, balance, and appreciation of the ecological commons in order to achieve true sustainability.

Conclusion

In conclusion, the relationship among technology, environment, and society is complex, deeply rooted, and misunderstood. We can now recognize social and environmental problems and attempt to come up with solutions, like sustainable development. The root of the problems, though, is deeply ingrained in Western philosophy about the natural world. We have mistakenly given technology priority and power over the environment and ourselves. The social and environmental problems and inequities that result are somehow accepted as part of the deal.

The way the perceive and think about the world around us is out of balance. We have lost our connection to each other and to the environment. The commons – the parts of life that are most fundamentally shared and connecting – are being forgotten and devalued. Commons' value is based on economic worth only, not for basic, intrinsic beauty, emotional, or spiritual "value"– things that make human nature just that. Human arrogance has reached a level where it is thought that engineered versions of nature and natural processes, even childbirth, will be superior to what has been created through millions of years of evolution. Bioengineering is already here to stay – we have changed the way food is grown and produced; we have altered and reconstructed this commons. Human engineering or atmospheric engineering may be close behind.

Western society needs help. The relationship needs help. It is dangerously out of balance and priorities are far off course. We are approaching a treacherous point on Earth where technology is valued over everything and seen as superior to all things natural, without exception. Humans are not skilled problem solvers as it is, and will carry this weakness and resultant inequalities

through all technological progress. We run a serious risk of self-destruction if we continue on this path without opening our eyes and tuning in. Fatality is not certain but it is distinctly possible. It might not be too late to re-evaluate and repair the technology-environment-society dilemma, but it is a major undertaking. To truly get to the root of the problem and get into balance, it will require an unraveling of what is normal to many people, the underlying philosophy guiding how and why we do what we do. If we do not dig deeper, our attempts to solve the repair list of social-environmental problems will be little more than temporary technological band-aids on a wound that requires major surgery.